

4. (a)	Arrows correct	1A
	Labels (tension / T , weight / Mg)	1A
	Or Either force correct and with label	1A
	All correct.	1A

As M accelerates upward,

$$T - Mg = Ma$$

i.e. $T = Mg + Ma$

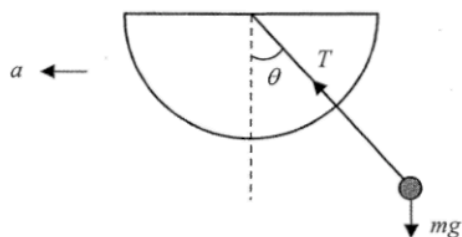
Greater tension needed ($T > \text{original tension} = Mg$) and thus the spring extends further.
i.e. M is closer to the bottom, smaller h in Figure 4.2

	1M
	1A
	4

(b)	Change in tension / reading = $2 \text{ N cm}^{-1} \times 0.5 \text{ cm} = 1 \text{ N}$	
	$\therefore T - Mg = 1 \text{ N}$ i.e. $T = 6 \text{ N}$	1A
	$T - Mg = Ma$	
	$1 \text{ N} = \frac{5 \text{ N}}{g} a$ (accept $M = 0.5 \text{ kg}$)	1M
	$a = \frac{1}{5}g$ or $0.2 g$ (upward) (or $\frac{g}{5} = 2 \text{ m s}^{-2}$ or 1.96 m s^{-2})	1A
		3

(c)	$a_y = -0.5g = -g \cos \theta$	1M
	$\therefore \theta = 60^\circ > 45^\circ$, 'Landscape display' results.	1A
	Or When the phone is rotated 45° clockwise,	
	$a_y = -g \cos 45^\circ$ or $-g \sin 45^\circ$	1M
	$= -\frac{g}{\sqrt{2}} = -0.71 g$ or -6.94 m s^{-2} (or -7.07 m s^{-2})	
	'Landscape display' results as $a_y = -0.5 g > -0.71 g$.	1A
	Or 'Landscape display' results as magnitude of $a_y < \text{that at } 45^\circ$	1A
		2

5. Diagram



1A

Tie one end of the string to the metal ball and the other end through the centre/hole of the protractor.

1A

When the train is at rest, held fixed the protractor in the plane along the direction of motion such that the string is on, say, the 90° mark.

1A

When the train is accelerating with acceleration a , the string will make an angle, say θ , with the vertical. Measure the angle θ .

1A

Let T be the tension of the string,

Vertically : $T \cos \theta = mg$ ①

1M

Horizontally : $T \sin \theta = ma$ ②

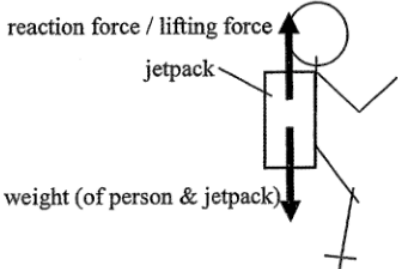
where m is the mass of the ball

$$\frac{\textcircled{2}}{\textcircled{1}} : \tan \theta = \frac{a}{g}$$

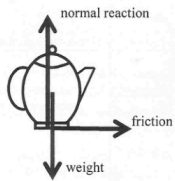
$$a = g \tan \theta$$

1A

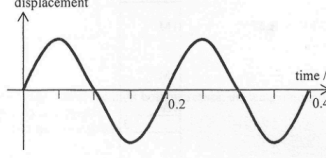
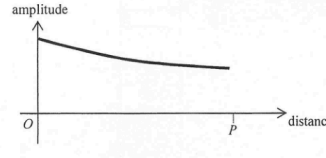
6

4. (a) According to Newton's second law of motion, a (net) force acts on the water so as to change its momentum. (Or magnitude of the force equals the rate of change of momentum of water).	1A	
According to Newton's third law of motion, a force acting downwards on the ejecting water (by the jetpack), the water exerts a reaction (equal but upward / opposite) on the jetpack / person as well.	1A 1A	
(b) 	3 1A	
(c) (i) $F = \frac{\Delta p}{\Delta t} = \frac{\Delta m}{\Delta t} \times (\vec{v} - \vec{u})$ $\frac{\Delta m}{\Delta t} \times (10 - (-10)) \text{ m s}^{-1} = 1000 \text{ N}$ $\frac{\Delta m}{\Delta t} = 50 \text{ (kg s}^{-1}\text{)}$	1 1M 1A	Accept using kg as the unit.
(ii) $\left(\frac{\Delta m}{\Delta t}\right)gh + \frac{1}{2}\left(\frac{\Delta m}{\Delta t}\right)v^2$ $= (50 \text{ kg s}^{-1})(9.81 \text{ m s}^{-2})(7.5 \text{ m}) + \frac{1}{2}(50 \text{ kg s}^{-1}) \times (10 \text{ m s}^{-1})^2$ $= 6178.75 \text{ W or } 6.17875 \text{ kW}$	2 1M 1M 1A	Accept using m for $\left(\frac{\Delta m}{\Delta t}\right)$ For $g = 10 \text{ m s}^{-2}$, $3750 \text{ W} + 2500 \text{ W} = 6250 \text{ W}$
(d) The same as the same lifting force / (water) jet speed is needed.	3 1A 1A 2	

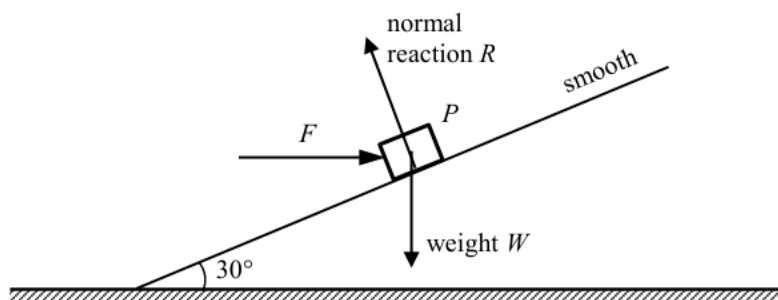
1. (a)	A larger bulb improves the sensitivity of the thermometer.	1A	
	<u>Or</u> A larger bulb minimizes the effect on the temperature reading due to the other parts of the thermometer stem that are exposed to different temperatures.	1A	
		1	
(b) (i)	$E = mc\Delta T$	1M	
	$= 0.015 \times (2.9 \times 10^3) \times (20 - 15)$	1A	
	$= 217.5 \text{ J}$	2	
(ii)	Time taken to reach air temperature = $\frac{217.5}{0.5}$	1M	
	$= 435 \text{ s}$	1A	
		2	

5. (a) 	1A+1A	
	2	
(b) $\omega = \pi \text{ s}^{-1}$ $F = m r \omega^2$ $= (1)(0.3)(\pi)^2$ $= 2.96 \text{ N (towards the centre of the turntable)}$	1A 1M 1A	
Alternative method: $v = 0.3\pi \text{ m s}^{-1}$ $F = m \frac{v^2}{r}$ $= 2.96 \text{ N}$	1A 1M 1A	
	3	
(c) The initial linear speed of the teapot $= r\omega = 0.3\pi \text{ m s}^{-1}$ Deceleration of the teapot $a = \frac{f}{m} = \frac{10}{1} = 10 \text{ m s}^{-2}$ Distance travelled s is given by $v^2 - u^2 = 2as$ $s = \frac{u^2}{2a} = \frac{(0.3\pi)^2}{2(10)}$ $= 0.044 \text{ m (or 4.4 cm)}$ Alternative method: The initial linear speed of the teapot $= r\omega = 0.3\pi \text{ m s}^{-1}$ K.E. of the teapot is dissipated as work done against friction $\frac{1}{2} mu^2 = fd$ $d = \frac{mu^2}{2f} = \frac{(1)(0.3\pi)^2}{2(10)}$ $= 0.044 \text{ m}$	1M 1M 1A 1M 1A	3

61

6. (a) (i) $v = f\lambda$ $= 5 \times 4$ $= 20 \text{ cm s}^{-1}$	1M 1A	2
(ii) Y is moving upwards at $t = 0$.	1A	1
(iii) 	1A 1M	2
(b) (i) The water waves from A and B are in anti-phase at Q. Or The path difference at Q $= (n + 1/2)\lambda$. Destructive interference occurs to form a minimum.	1M 1M 1A	2
(ii) Path difference at Q $= 1.5\lambda = 3 \text{ cm}$ $\lambda = 2 \text{ cm}$	1M 1A	2
(iii) 	1A	1

5. (a) (i)



1A

1A

Forces R (or N) and W (or Mg) correctly indicated and labelled.

[MR:

1A for weight with correct label (W/Mg /weight/gravity)1A for normal reaction with correct label (R/N /normal force/contact force/reaction)1A only: if both W and R drawn (without/wrong label)

Withhold 1A for additional force drawn (more than 2 forces)

- W components added with correct labels (ignored)- W components added without correct labels (treated as additional force drawn)

2

(ii) $R \cos 30^\circ = W = Mg$ and $R \sin 30^\circ = F$

$$F = Mg \times \tan 30^\circ = 56.63806 \text{ N} \approx 56.6 \text{ N}$$

$$N = R = \frac{Mg}{\cos 30^\circ} = 113.27612 \text{ N} \approx 113 \text{ N}$$

Or $R = W \cos 30^\circ + F \sin 30^\circ$ and $W \sin 30^\circ = F \cos 30^\circ$

1M

 $M = 10 \text{ kg}$; $Mg = 98.1 \text{ N}$

[MR: 1M for both equations are correct]

1A

For $g = 10 \text{ m s}^{-2}$, $57.735027 \text{ N} \approx 57.7 \text{ N}$

1A

 $115.470054 \text{ N} \approx 115 \text{ N}$

(1M)

[MR: 1M for both equations are correct]

3

(b) (i) $g \sin \theta = 9.81 \sin 30^\circ = 4.905 \text{ m s}^{-2} \approx 4.91 \text{ m s}^{-2}$

1A

 5 m s^{-2} for $g = 10 \text{ m s}^{-2}$,

1

(ii) Decrease

as the component of F perpendicular to the incline no longer acts on the block / inclineOr as F is removed, the force pressing the block / incline would decrease (only the weight's component left)

1A

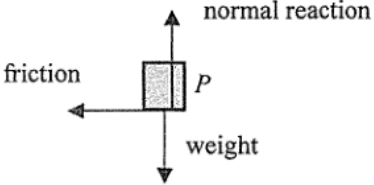
1A

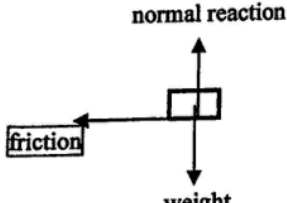
		Answer		
		De	In / Un	NA
Reason	✓	2	0	1
	✗ / NA	0	0	0
	Not complete	1	0	0

2

NA – Not answered / Un - Unchanged

3. (a) According to Newton's third law of motion, force on air streams and the thrust on the quadcopter are (equal and) opposite, with magnitude given by $F = \text{rate of change of momentum (of air streams)}$. Therefore the thrust (upward) on the quadcopter balances its weight for a certain speed of air streams.	1A	
	1A	
	2	
(b) (i) Volume of air streams propelled in 1 s $V = 0.284 \text{ m}^3$ $m_a = \text{density} \times \text{volume}$ $= 1.20 \times 0.284 \text{ m}^3 = 0.3408 \text{ kg}$	1M	
	1A	
	2	
(ii) Weight of quadcopter = rate of change of momentum of air streams $1.38 \text{ kg} = m_a v - 0$ $1.38 \times 9.81 = (0.3408 \text{ kg}) \times v \quad (\text{From (b)(i)})$ $v^2 = 39.723592$ $v = 6.302665 \text{ m s}^{-1} \approx 6.30 \text{ m s}^{-1}$	1M	
	1A	For $g = 10 \text{ m s}^{-2}$, $v = 6.363408 \text{ m s}^{-1} \approx 6.36 \text{ m s}^{-1}$
	2	
(c) (i) 	2A	
	2	
(ii) centripetal force required $F = \frac{1.38 \times 15^2}{50}$ $= 6.21 \text{ N}$	1M	
	1A	
	2	
(iii) Let $T = \text{total thrust on the quadcopter}$ Vertical: $T \cos \theta = 1.38 \text{ kg} \quad (g = 9.81 \text{ m s}^{-2})$ Horizontal: $T \sin \theta = 6.21 \text{ N} \quad (\text{From (c)(ii)})$ } On solving the two equations, $\tan \theta = 0.458716$ $\theta = 24.641662^\circ \approx 24.6^\circ$	1M	Accept $\tan \theta = \frac{\text{centripetal force}}{\text{weight}}$
	1A	For $g = 10 \text{ m s}^{-2}$, $\tan \theta = 0.45$ $\theta = 24.227745^\circ \approx 24.2^\circ$
	2	

3. (a) By $Fs = \frac{1}{2}mv^2$, $0.30 \times 0.75 = \frac{1}{2} (0.20) v^2$ $v = 1.5 \text{ m s}^{-1}$	1M 1A	Or $v^2 = u^2 + 2as$ and $F = ma$ $v^2 = 0^2 + 2\left(\frac{0.30}{0.20}\right)(0.75)$
(b) 	2A	Accept: normal force, gravity
(c) Height = $\frac{1}{2}gt^2 = \frac{1}{2}(9.81)(0.35)^2$ $= 0.6008625 \text{ m} \approx 0.601 \text{ m}$	1M 1A	Accept: 0.60 m ~ 0.613 m

4. (a) $\omega = \frac{2\pi}{60}$ $= \frac{\pi}{30}$ or $0.105 \text{ (rad s}^{-1}\text{)}$	1A	
(b) 	1A	Accept: adhesive force
(c) The required centripetal force $F_c = m\omega^2 r$, where r is the distance of plasticine Q from the clock's centre. As m and ω are constant, smaller r gives smaller centripetal force F_c. The claim is correct.	1A+1A	